

KAYES, Mali

WMO: 612570

Lat: 14.482N

Lon: 11.399W

Elev: 169

StdP: 14.61

Time Zone: 0.00 (E00)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	60.8	63.4	18.3	13.9	80.2	22.7	17.2	82.0	17.5	82.5	16.4	80.9	4.5	120	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	19.6	110.7	72.7	109.1	72.4	107.4	72.3	82.5	92.6	81.8	91.7	81.0	90.8	7.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	157.1	87.0	79.3	153.2	86.6	78.7	149.9	86.2	46.5	92.7	45.6	92.1	44.8	91.0	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.8	15.0	13.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			44.0	85.2	1.9	1.3	42.7	86.2	41.6	87.0	40.5	87.7	39.2	88.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	86.7	77.7	82.7	89.9	94.8	97.7	93.8	85.5	83.5	84.0	86.9	84.1	79.5
	DBStd	6.97	3.66	4.51	4.44	3.56	3.15	4.43	3.63	3.21	2.63	2.65	2.93	4.28
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	13385	858	915	1237	1345	1478	1313	1101	1038	1020	1143	1022	915
	CDD65	7910	393	495	772	895	1013	863	636	573	570	678	572	450
	CDH74	113262	4470	6457	11993	15166	17480	13981	8449	6802	6835	9015	7364	5250
	CDH80	70418	2267	3751	7975	10907	13029	9716	4378	2993	3165	5045	4403	2789
Wind	WSAvg	6.9	7.0	7.5	6.7	6.9	7.7	8.9	7.9	6.4	5.6	5.4	5.4	6.9
Precipitation	PrecAvg	24.8	0.0	0.0	0.0	0.0	0.5	2.9	6.0	8.7	5.2	1.6	0.0	0.0
	PrecMax	35.0	0.1	0.6	0.2	0.2	1.6	6.7	13.4	16.0	10.6	8.0	0.8	0.4
	PrecMin	16.9	0.0	0.0	0.0	0.0	0.0	0.0	0.2	3.1	2.3	0.0	0.0	0.0
	PrecStd	4.6	0.0	0.1	0.0	0.0	0.4	1.5	2.8	3.2	1.9	1.4	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.1	105.4	109.8	112.4	113.0	110.2	101.4	98.6	96.8	102.8	103.1	101.7
		MCWB	64.6	68.6	69.7	73.7	73.2	74.4	79.6	78.8	80.8	74.0	69.5	67.9
	2%	DB	95.6	102.2	107.2	110.6	111.0	107.6	98.0	94.9	94.6	100.4	101.3	98.7
		MCWB	63.6	67.0	69.5	72.5	73.1	75.0	79.3	79.8	81.0	75.3	69.2	66.5
	5%	DB	93.4	99.6	105.3	109.0	109.3	105.3	95.7	92.5	93.2	98.6	99.6	96.0
		MCWB	62.7	65.8	68.7	71.7	73.0	75.7	79.5	80.2	81.1	77.3	69.1	65.1
	10%	DB	91.2	96.9	103.3	107.0	107.5	103.2	93.7	90.6	91.6	96.7	97.6	93.2
		MCWB	61.6	64.3	67.8	71.0	73.4	76.1	79.3	79.9	80.6	77.5	68.7	64.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	71.3	75.7	79.4	79.9	81.3	81.9	82.8	83.7	83.4	78.1	71.7
		MCDB	90.9	99.6	104.1	107.2	103.1	98.0	95.0	90.8	91.7	94.5	88.9	96.1
	2%	WB	66.3	69.4	73.0	76.4	78.8	80.0	81.0	81.8	82.7	82.0	75.9	68.9
		MCDB	91.6	98.5	100.9	103.3	100.9	96.4	92.7	90.0	91.4	93.7	88.0	93.4
	5%	WB	64.4	67.5	71.0	74.9	77.8	79.2	80.5	81.0	81.9	81.2	73.4	66.6
		MCDB	89.2	96.3	100.6	101.6	99.6	95.2	91.4	89.1	90.4	92.5	89.3	91.9
	10%	WB	62.7	65.6	69.3	73.6	76.8	78.5	79.9	80.3	81.2	80.4	71.2	64.9
		MCDB	87.6	93.9	100.1	101.8	98.6	94.5	90.4	88.3	89.1	91.3	91.9	90.9
Mean Daily Temperature Range	5% DB	MDBR	27.2	26.9	26.6	24.5	19.6	17.5	14.1	13.1	15.0	21.1	29.3	27.3
		MCDBR	30.6	30.0	28.3	25.9	22.2	19.5	16.6	15.4	16.7	23.4	31.8	30.9
		MCWBR	12.7	11.8	10.8	9.6	8.1	5.5	4.7	5.3	6.2	7.0	10.4	11.6
	5% WB	MCDBR	28.7	29.2	25.5	23.0	18.3	17.6	14.7	14.1	15.7	19.7	27.5	28.5
		MCWBR	12.8	12.0	10.9	9.7	6.7	5.4	5.1	6.2	6.9	7.3	10.3	11.4
Clear-Sky Solar Irradiance	taub		0.515	0.552	0.656	0.662	0.699	0.727	0.653	0.584	0.576	0.596	0.511	0.485
	taud		1.896	1.806	1.600	1.634	1.586	1.598	1.763	1.924	1.919	1.855	1.967	1.997
	Ebn at Noon		246	244	223	221	210	201	218	236	239	230	246	251
	Edh at Noon		61	69	87	84	87	84	72	62	62	65	56	53
All-Sky Solar Radiation	RadAvg		1692	1887	2096	2207	2145	1986	1829	1724	1833	1846	1724	1641
	RadStd		104	95	99	83	88	72	92	78	76	40	73	67

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page